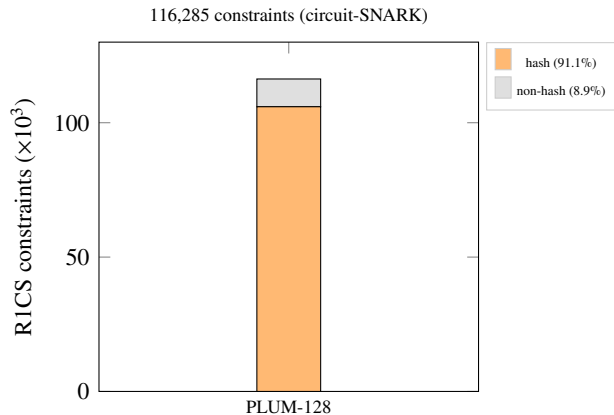
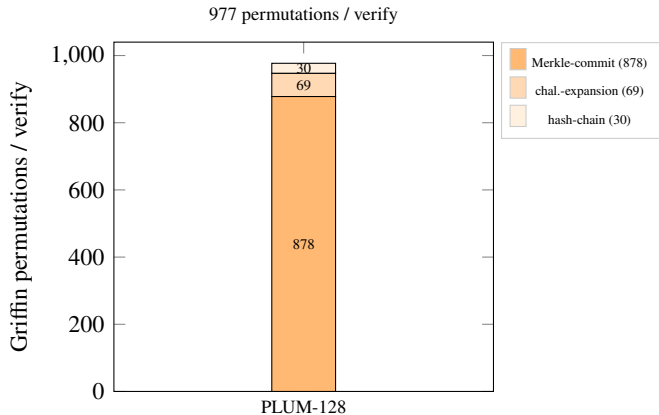




*PLUM-128 verification is hash-dominated (PLUM §4.2): the Griffin permutation is the load-bearing precompile target. Constraint counts are circuit-SNARK R1CS, not zkVM cycles; the zkVM cost share is measured separately in the evaluation.*